

< [vercel app link](#) >

< [github repo link](#)>

(my) SUMMARY

Brief Overview: For this version, I focused on research. I pulled policy resources from the UN, OECD, and the USGS. I looked at academic sources, and I read materials from advocacy organizations like Friends of the Congo. A small repository of my research materials as well as a working brief I am preparing can be found in the [research/](#) folder of my repository. I reworked the [globe](#) page to be more accurate-and more beautiful-and I added interactive elements to the model-select page to keep a somatic and immersive feel to the webpage. Finally, I researched and developed a hand-drawn diagram and svg for the Samsung s25.

Challenges:

- Developing component diagrams for the Samsung and Pixel models is proving more complex. I'm screenshotting teardown videos and drawing from there, but it is much less accessible than the iPhone 16.
- Getting a working horizontal-scroll for my model-select page has been tricky. It still needs polish -- I don't like the cranberry halo, or the size of the panel that is highlighted on hover.
- Research still required! Several of the pages point to "rare earths" pipelines, which is a bucket for rare earth elements. I need to learn more deeply about specific mineral pipelines and pathways. Considering using UN/OECD APIs for final version.
- Polish is a never-ending project, but there are lots of things I want to try: adding a more human element with quotes/images in the globe page, tweaking hover effects, changing buttons and transitions.

CLAUDE Summary

We started the session by auditing the existing build against the Version 3 checklist, confirming the carousel and Samsung/Pixel diagram stubs were already in place, and identifying the info cards as the only incomplete task. Working from the research brief PDF, we wrote 4 cards per mineral across all 6 minerals (cobalt, tantalum, tungsten, gold, tin, rare-earths), grounding each in real sourced facts — DRC's 60% cobalt supply, 15/19 Chinese-owned concessions, the coltan exploitation chain through Rwanda and Uganda, the Washington Accord signed in June 2025, China refining 90% of the world's rare earth elements, and Belt & Road's \$57B mineral investment footprint. Each card follows the same editorial structure: The Mine / The Metal (or a component-specific angle) / The Journey / What Can Change.

From there we tackled the globe, which was rendering as a blank beige sphere. The root cause was a two-layer problem: SVG files make terrible WebGL textures (they're rasterized at low resolution and look grainy), and the CSP header connect-src 'self' was

silently blocking the GeoJSON fetch needed to draw country polygons. We fixed both — replacing the SVG atlas with a 2×2 cream canvas pixel as the ocean texture, enabling `showGraticules` for a cartographic parchment look, loading Natural Earth country polygons for ink-outlined landmasses, and adding `raw.githubusercontent.com` to `connect-src`. We also redesigned the page layout: the supply chain pipeline and mine location moved from the bottom to a top chrome bar with a cream gradient fade, pipeline nodes became clickable buttons that rotate the globe to each step's coordinates with a 1.1s animated pan, and past steps trail in cranberry. The card panel got darkened body text, enlarged Playfair titles, cranberry mono citations, and rectangular PREV/NEXT buttons.

The select page carousel went through the most debugging. We replaced the broken static layout with a proper `objectFit: cover` crop using derived container dimensions — the key insight being that $CH/CW = (1600 \times 3) / 2150$ forces the browser's height-driven cover scale to make the rendered SVG exactly 3 card-widths wide, so `objectPosition` values of 0% / 50% / 100% crop each phone precisely. We added snap-swipe drag (pointer events, `dragDistRef` for stale-closure safety, 30px threshold), a zoom-in click transition that scales the phone to `scale(9)` while a cream overlay fades over the page, and a soft ambient cranberry glow on hover. The click-through was broken entirely by `setPointerCapture` — it redirects synthesized click events to the container element, preventing child button `onClick` from ever firing — and was fixed by removing it.

Finally, we added the Samsung Galaxy S25 component diagram by embedding the PNG as a base64 inside an SVG wrapper (lossless, no vectorization artifacts), refactored `LABEL_Y` into a per-phone nested object so each device can have independent label positions, and shipped a final polish pass: a `pageEnter` fade+rise transition on every route, a fully redesigned globe info card with cranberry rule accents, italic Playfair headings, pill dot-nav with a card counter, and a more dramatic active pipeline node with a cranberry glow halo.